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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The characterisation of m-pseudo-coherent complexes, and the passage from all m to pseudo-coherent. Lemma \ref{lemma-recognize-pseudo-coherent} and Lemma \ref{lemma-pseudo-coherent}, cited at proof lines 16267 and 16279. Give the base case of the induction and the pseudo-coherent conclusion.
- The distinguished triangle relating a truncation of a complex, the complex, and its top cohomology shifted. Derived Categories, Remark \ref{derived-remark-truncation-distinguished-triangle}, cited at proof lines 16270-16272. Supplies the triangle to which the cone lemma is applied.
- Cone lemma for pseudo-coherence in a distinguished triangle. Lemma \ref{lemma-cone-pseudo-coherent}, cited at proof lines 16273-16275. Reduces the claim to the truncation once the shifted cohomology module is known to be pseudo-coherent.

Source reference. Original lemma 066B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7930–7930. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 064W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7880–7880. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08J5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10838–10838. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7875–7875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7878–7878. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $m \in \mathbf{Z}$. Let $K^\bullet \in D^-(R)$ such that $H^i(K^\bullet)$ is $(m-i)$-pseudo-coherent (resp. pseudo-coherent) for all i. Then $K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent).*

Proof. Assume $K^\bullet$ is an object of $D^-(R)$ such that each $H^i(K^\bullet)$ is $(m-i)$-pseudo-coherent. Let n be the largest integer such that $H^n(K^\bullet)$ is nonzero. We will prove the lemma by induction on n. If $n < m$, then $K^\bullet$ is m-pseudo-coherent by Lemma 064W. If $n \geq m$, then we have the distinguished triangle

$$(\tau_{\leq n-1}K^\bullet, K^\bullet, H^n(K^\bullet)[-n])$$

(Derived Categories, Remark 08J5) Since $H^n(K^\bullet)[-n]$ is m-pseudo-coherent by assumption, we can use Lemma 064R to see that it suffices to prove that $\tau_{\leq n-1}K^\bullet$ is m-pseudo-coherent. By induction on n we win. (The pseudo-coherent case follows from this and Lemma 064U.) □